

Sotility Ecosystem: Smart Contract Overview and Architecture

Smart Contract Overview

TOKEN LAYER

1. **SotilityStableToken (SST)**

- Revenue-backed stablecoin (1:1 USD peg)
- Minted only against verified business revenue
- IPFS receipt tracking for transparent auditability
- Only approved minters (SotilityExchange) can create tokens

2. **SotilityOwnershipToken (SOT)**

- Equity token representing ownership in the Sotility ecosystem
- Dividend-eligible from protocol revenue
- Governance rights for protocol-level decisions
- Initial supply: 1 billion tokens with predefined distribution

3. **SoGoodUtilityGovernance (SUG)**

- Utility and governance token for the social platform
- Time-locked to encourage long-term commitment
- Earned through verified contributions to the ecosystem
- Merit-based reward allocation with algorithmic distribution

4. **SotilityVeToken (veSOT)**

- Vote-escrowed token implementation for SOT holders
- Allows locking SOT for enhanced governance weight (up to 2x)
- Linear decay mechanism for voting power as lock time decreases
- Maximum lock time: 4 years

5. **SotilityLiquidStaking (lstSOT)**

- Enables staking SOT while maintaining liquidity
- Issues lstSOT tokens representing staked SOT plus accrued rewards
- Dynamic exchange rate based on total rewards in the system

- Maintains partial governance rights (0.8x multiplier)

SOCIAL LAYER

1. SoGoodFeed

- On-chain registry for social utility posts
- AI-verified tipping mechanism
- IPFS content storage with on-chain references
- Tracks contribution and engagement history

2. SotilityProfileRegistry

- Stores reputation metrics for ecosystem participants
- Tracks Sotilitarian Status (long-term) and Utilitier Level/Score (short-term)
- Four-tier system: Bronze, Silver, Gold, Platinum
- Provides eligibility for rewards and governance boosts

3. SotilityBadgeNFT

- Soulbound NFT badges for achievements and status
- Non-transferable credentials for expertise and contribution
- IPFS metadata for detailed achievement verification
- Revocable by authorized minters if needed

4. SotilityContentGovernance

- Decentralized content moderation with economic incentives
- Stake-based moderation system with rewards for consensus
- Slashing mechanism for misaligned votes
- Escalation and dispute resolution framework

FINANCIAL LAYER

1. SotilityExchange

- Business onboarding and verification platform
- Revenue proof submission and verification
- SST issuance against verified revenue
- Maintains registry of approved businesses

2. SotilityTreasuryRouter

- Routes protocol revenue to different ecosystem components

- Configurable allocation percentages (default: 40% SOT dividends, 40% SST reserves, 20% SUG campaigns)
- Event-driven for transparent tracking
- Updateable allocation targets for flexibility

3. **SotilityYieldEngine**

- AI-managed yield strategies for protocol assets
- Laddered leverage and risk-adjusted allocation
- Strategy management for maximizing returns
- Integration with external DeFi protocols

4. **SotilityVaultFactory**

- Deploys isolated SotilityYieldEngine instances
- Business-specific yield strategies
- IPFS metadata storage for strategy documentation
- Registry for tracking all deployed vaults

5. **SotilityBridgeAdapter**

- Multi-chain token bridging facility
- Chain registry with destination bridging contracts
- AI-enhanced fraud detection for suspicious transactions
- Event emission for relayer pickup

6. **SotilityAdaptiveTokenomics**

- Automatically adjusts tokenomic parameters based on ecosystem health
- Monitors user growth, retention, transaction value, content creation, and TVL
- Adapts inflation rates, reward amounts, and economic parameters
- Historical tracking for transparency and analysis

DAO LAYER

1. **ProposalManager**

- On-chain governance mechanism for protocol decisions
- Time-based quorum and voting system
- Multi-stage process (creation, signal voting, formal vote, execution)
- IPFS integration for detailed proposal documentation

2. **SoGoodDAOFactory**

- Deploys dedicated governance structures for communities
- Customizable governance parameters
- Enables localized decision-making
- Integration with main protocol governance

AI & SECURITY LAYER

1. **AIOracleManager**

- Registry for verified AI oracle agents
- Metadata tracking for version and capabilities
- Access control for oracle activation/deactivation
- Used by other contracts for verification requests

2. **MultiOracleAggregator**

- Combines results from multiple AI oracles
- Deviation thresholding for consensus
- Sybil-resistant aggregation
- Prevents oracle manipulation attacks

3. **AutonomousSoGoodAgent**

- Off-chain AI agent for content evaluation
- Monitors blockchain events
- OpenAI integration for scoring
- IPFS upload of evaluation results
- Automatic on-chain transaction execution

4. **SotilityEmergencyShutdown**

- Circuit breaker for security incidents
- Timelocked emergency activation
- Guardian committee control
- Granular contract pausing capabilities

IDENTITY & REPUTATION LAYER

1. **SotilityZKIdentity**

- Zero-knowledge proofs of reputation and credentials

- Privacy-preserving verification
- Selective disclosure capabilities
- Category-based proof verification

2. **SotilityCrossChainIdentity**

- Aggregates identity and reputation from multiple blockchains
- Weighted reputation scoring
- Bridge-based verification
- Chain-agnostic identity portability

3. **SotilityProofOfPersonhood**

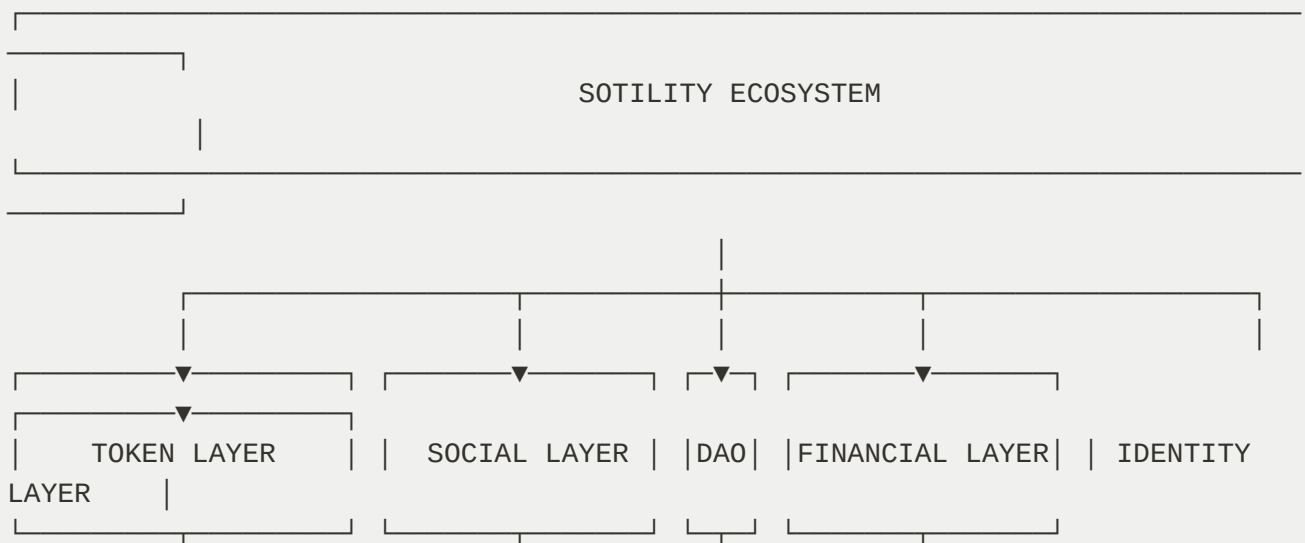
- Sybil-resistant human verification
- Time-limited credentials with renewal
- Revocation capabilities for fraudulent accounts
- Privacy-preserving implementation

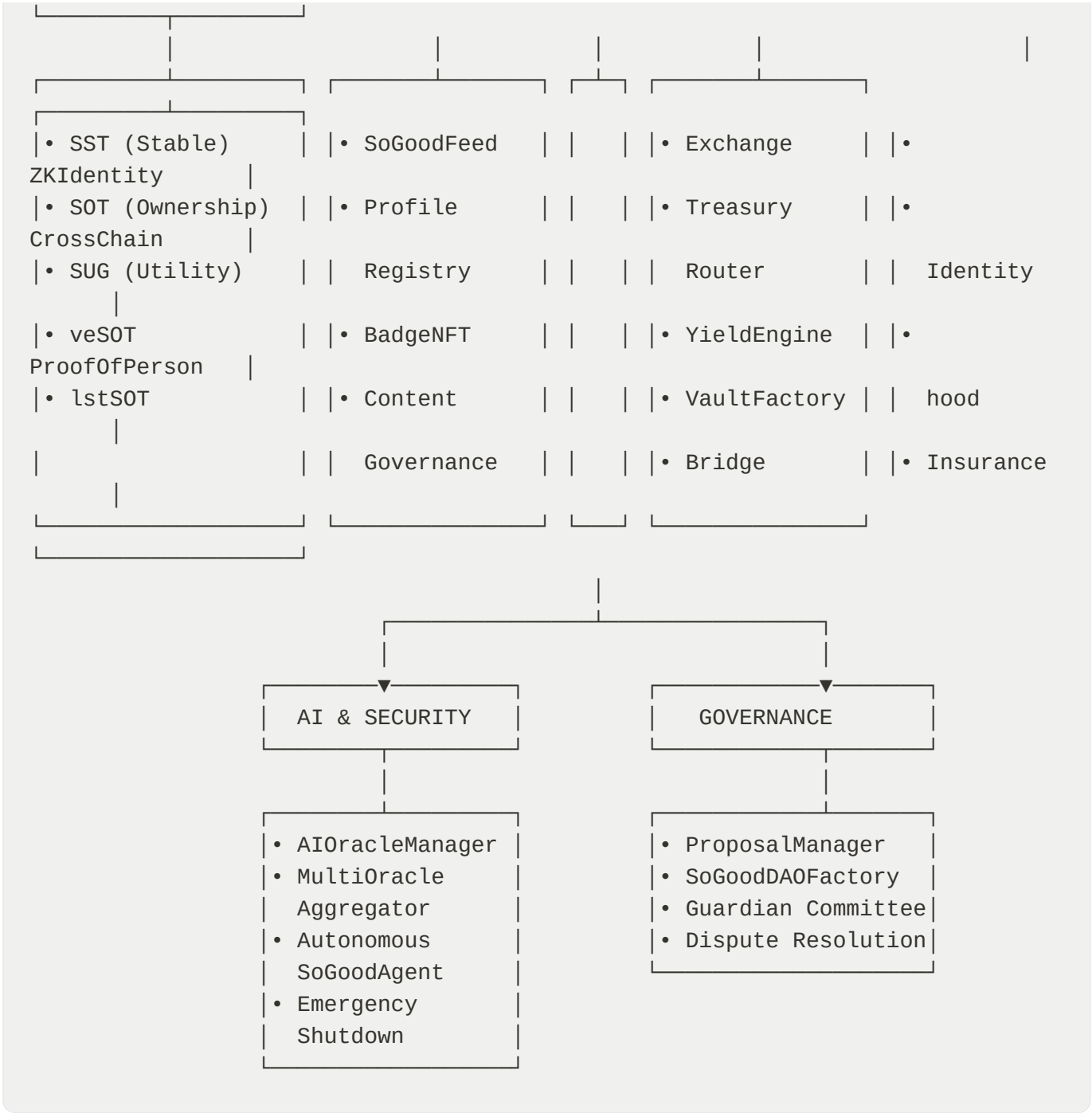
4. **SotilityInsurance**

- Protocol insurance against exploits and failures
- Policy-based coverage for ecosystem participants
- Claim submission and processing workflow
- Premium adjustments based on governance token holdings

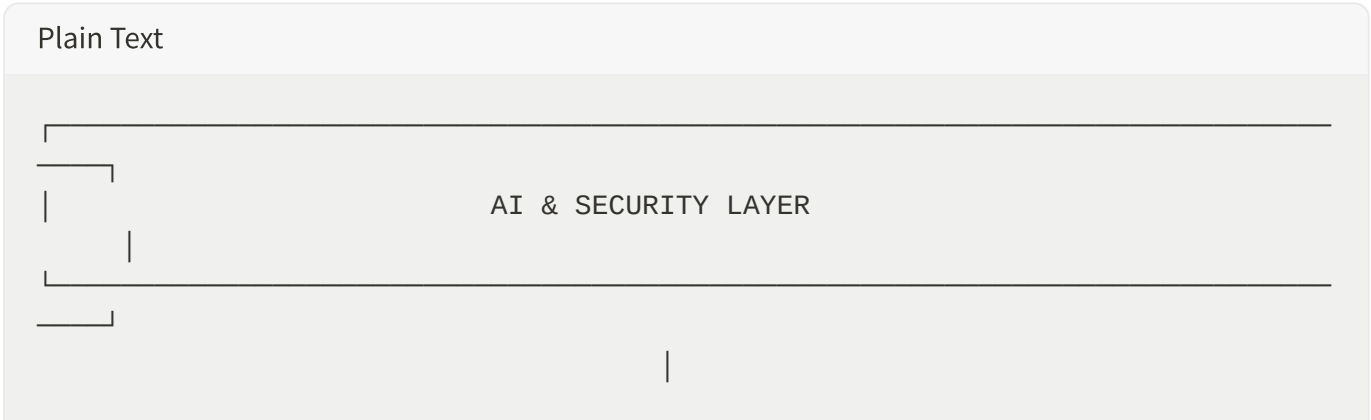
Sotility Ecosystem Architecture

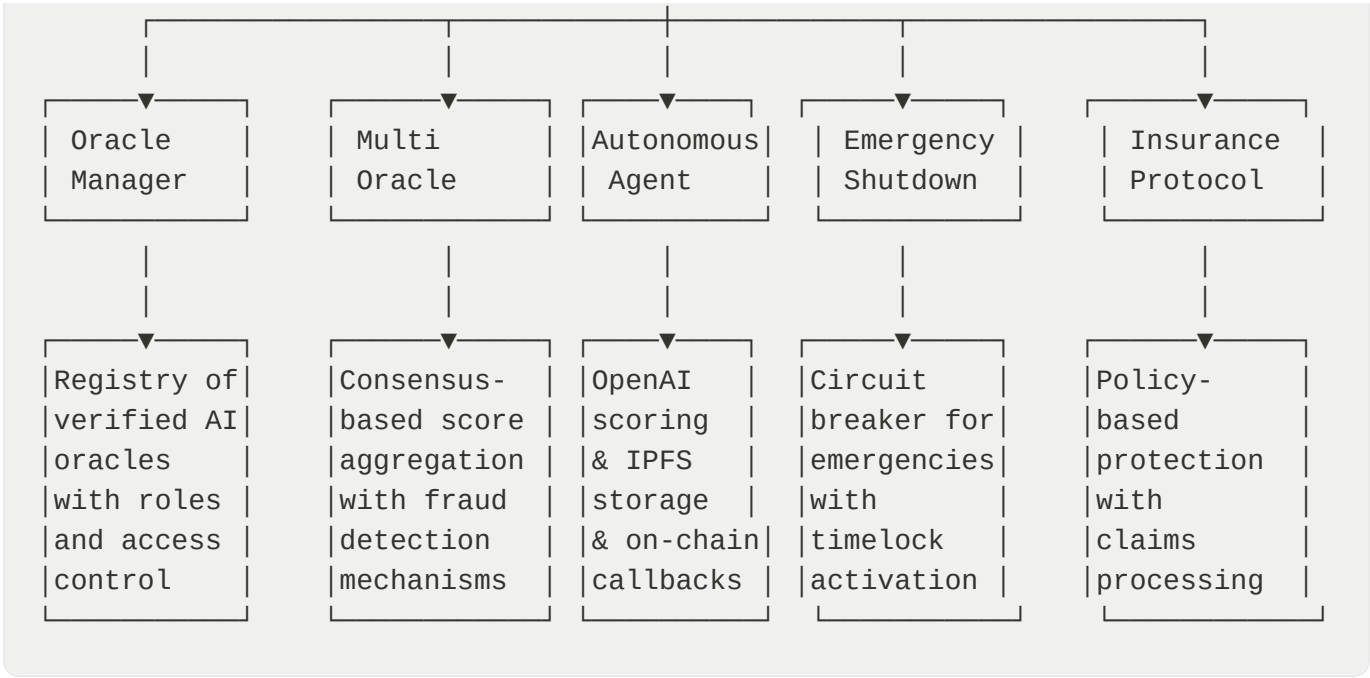
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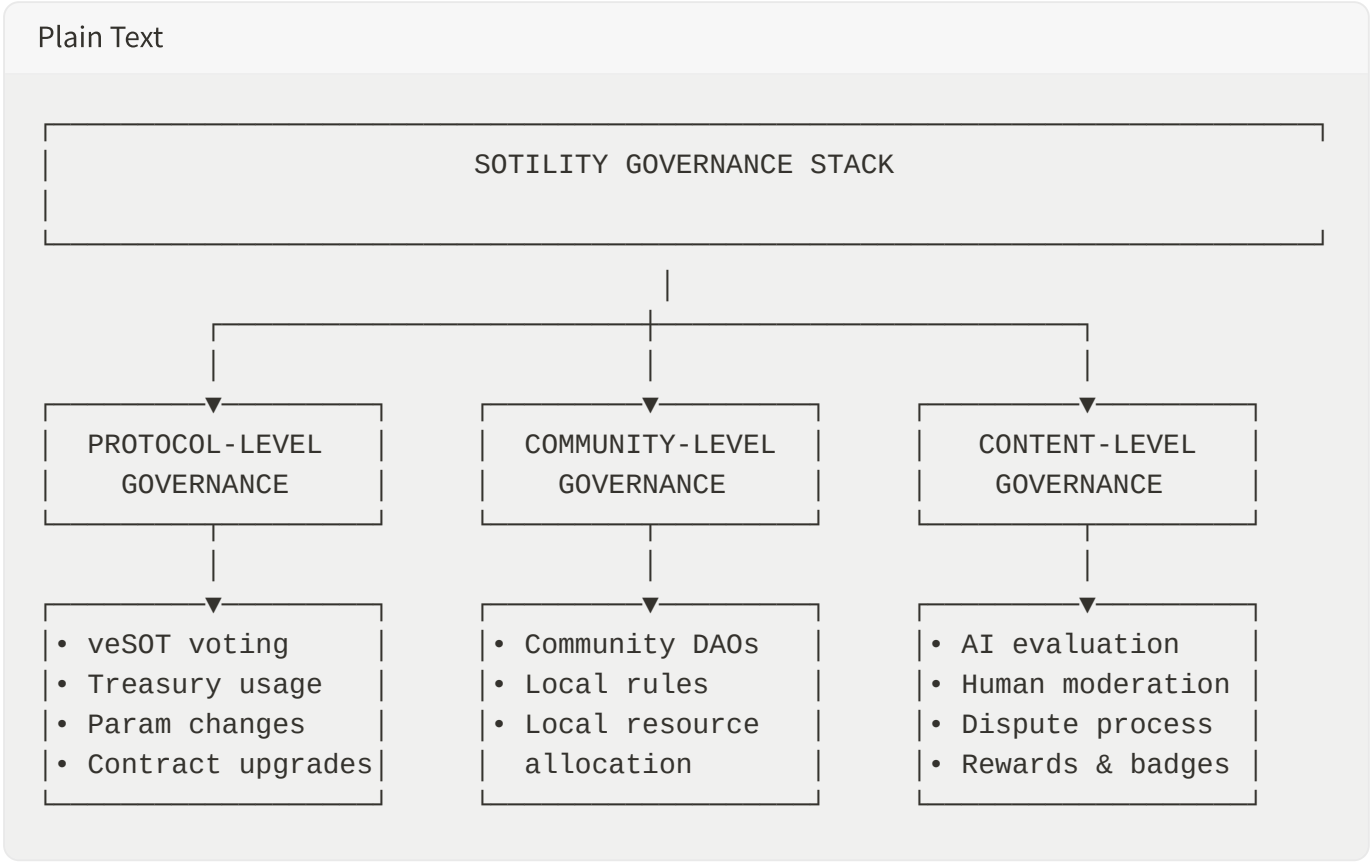


AI & Security Architecture

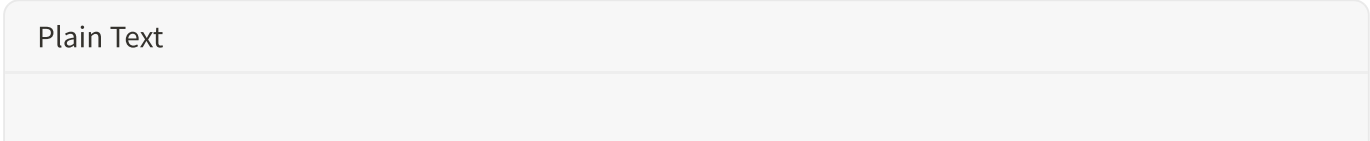


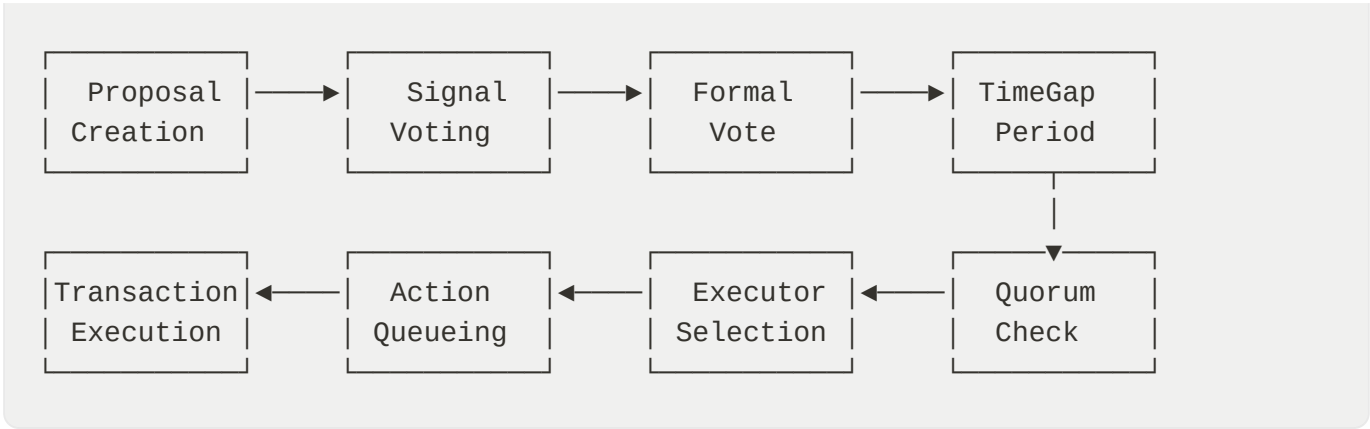


Governance Framework



Protocol-Level Governance Flow





Token Distribution Model

SOT (SotilityOwnershipToken)

Allocation Category	Percentage	Amount (Millions)	Vesting Period
Treasury	20%	200	Strategic releases by DAO vote
Team & Early Contributors	15%	150	2-year cliff, 4-year linear vesting
Ecosystem Development	25%	250	Quarterly releases over 5 years
Initial Community Sale	10%	100	Immediate (liquid)
Seed & Strategic Investors	15%	150	1-year cliff, 3-year linear vesting
Liquidity Provision	10%	100	80% initially locked for 2 years
Advisors & Partners	5%	50	1-year cliff, 2-year linear vesting
Total	100%	1,000	

SUG (SoGoodUtilityGovernance)

Allocation Category	Percentage	Amount (Millions)	Distribution Schedule
Content Rewards	50%	125	Continuous algorithmic distribution
DAO Treasury	20%	50	Controlled by governance
Community Airdrops	10%	25	Quarterly merit-based distributions
Platform Development	15%	37.5	3-year quarterly release
Early Adopter Incentives	5%	12.5	First-year campaign rewards
Total	100%	250	

Protocol Revenue Allocation

Allocation	Percentage	Purpose
SOT Dividends	40%	Distributed to SOT holders
SST Reserves	40%	Backs SST stability and new mints
SUG Campaigns	20%	Funds ecosystem growth and rewards

Key Features & Innovations

- AI-Powered Verification:** Autonomous AI agents that verify content quality and business revenue without requiring user understanding of blockchain technology.
- Multi-Token Economic Model:** Three-tier system (SOT, SUG, SST) creates separation of concerns while enabling powerful cross-token interactions.
- Reputation-Enhanced Economics:** Financial activities are weighted by reputation, creating meritocratic incentives aligned with positive social impact.

4. **Privacy-Preserving Identity:** Zero-knowledge proofs enable credential verification without revealing underlying data.
5. **Adaptive Tokenomics:** Automatically adjusts economic parameters based on ecosystem health metrics, creating counter-cyclical stability.
6. **Multi-Layered Governance:** Three interconnected governance layers (protocol, community, content) with appropriate mechanisms for each.
7. **Cross-Chain Identity Graph:** Unified reputation aggregation across multiple blockchains creates portable trust.
8. **Revenue-Backed Stability:** SST stablecoin backed by verified business revenue rather than traditional collateral.
9. **Liquid Governance:** Vote-escrowed and liquid staking mechanics enable capital efficiency without sacrificing governance rights.
10. **Progressive Decentralization Path:** Clear roadmap from centralized bootstrapping to fully decentralized community ownership.